

Trigonometry

ANSWER KEY



Right-triangle trigonometric ratios

- 1 In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin \theta$ and the length of the adjacent side.

$$\text{Adjacent} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12. \sin \theta = \frac{5}{13}.$$

- 2 In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos \theta$.

$$\text{Adjacent} = \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8. \cos \theta = \frac{8}{17}.$$

- 3 In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.

$$\text{Opposite} = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24. \sin \theta = \frac{24}{25}, \tan \theta = \frac{24}{7}.$$

- 4 In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin \theta$ and $\cos \theta$ for the angle θ opposite the side of length 20.

$$\sin \theta = \frac{20}{29}, \cos \theta = \frac{21}{29}.$$

Special angles

- 5 State the exact value of $\sin(30^\circ)$.

$$\sin(30^\circ) = \frac{1}{2}.$$

- 6 State the exact value of $\cos(45^\circ)$.

$$\cos(45^\circ) = \frac{\sqrt{2}}{2}.$$

- 7 State the exact value of $\tan(60^\circ)$.

$$\tan(60^\circ) = \sqrt{3}.$$

- 8 State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

$$\sin(90^\circ) = 1 \text{ and } \cos(0^\circ) = 1.$$

Converting between radians and degrees

- 9 Convert 180° to radians.

$$180^\circ = \pi \text{ radians}.$$

- 10 Convert 90° to radians.

$$90^\circ = \frac{\pi}{2} \text{ radians.}$$

- 11 Convert $\frac{5\pi}{6}$ radians to degrees.

$$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ.$$

- 12 Convert 270° to radians.

$$270^\circ = \frac{3\pi}{2} \text{ radians.}$$

The Pythagorean identity

- 13 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

$$\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{4}{5}.$$

- 14 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$.

$$\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{12}{13}.$$

- 15 Given $\sin \theta = 0.6$, find $\cos^2 \theta$.

$$\cos^2 \theta = 1 - 0.36 = 0.64.$$

Simplifying trigonometric expressions

- 16 Simplify $\sin^2 \theta + \cos^2 \theta + 1$.

$$\sin^2 \theta + \cos^2 \theta = 1, \text{ so the expression equals } 1 + 1 = 2.$$

- 17 Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$ (for $\sin \theta \neq 0$).

$$1 - \cos^2 \theta = \sin^2 \theta, \text{ so the expression equals } \frac{\sin^2 \theta}{\sin \theta} = \sin \theta.$$

- 18 Simplify $\tan \theta \cos \theta$ (for $\cos \theta \neq 0$).

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \text{ so } \tan \theta \cos \theta = \sin \theta.$$

Graphs: amplitude and period

- 19 State the amplitude and period of $y = 5\sin(x)$.

Amplitude: 5. Period: 2π .

- 20 State the amplitude and period of $y = 3\cos(2x)$.

Amplitude: 3. Period: $\frac{2\pi}{2} = \pi$.

- 21 State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.

Amplitude: 2. Period: $\frac{2\pi}{1/3} = 6\pi$.

- 22 A sine graph has amplitude 4 and period π . Write its equation in the form $y = A\sin(Bx)$.

$A = 4$. $B = \frac{2\pi}{\pi} = 2$. So $y = 4\sin(2x)$.

Graphs: phase shift and vertical shift

- 23 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.

$$y = \sin\left(x - \frac{\pi}{3}\right) + 2.$$

- 24 State the vertical shift and midline of $y = 3\cos(x) + 5$.

Vertical shift: up 5. Midline: $y = 5$.

- 25 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.

$$y = \cos\left(x + \frac{\pi}{4}\right).$$

Solving basic trigonometric equations

- 26 Solve $2\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

$$\sin\theta = \frac{1}{2}. \text{ Solutions: } \theta = 30^\circ \text{ or } \theta = 150^\circ.$$

- 27 Solve $\sqrt{2}\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

$$\cos\theta = \frac{\sqrt{2}}{2}. \text{ Solutions: } \theta = 45^\circ \text{ or } \theta = 315^\circ.$$

- 28 Solve $\tan\theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

Solutions: $\theta = 135^\circ$ or $\theta = 315^\circ$.

- 29 Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

$$\cos\theta = -\frac{1}{2}. \text{ Solutions: } \theta = 120^\circ \text{ or } \theta = 240^\circ.$$

Reference angles in other quadrants

- 30** Find $\sin(150^\circ)$ using the reference angle.

The reference angle is 30° ; in Quadrant II, sine is positive. $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$.

- 31** Find $\cos(240^\circ)$ using the reference angle.

The reference angle is 60° ; in Quadrant III, cosine is negative. $\cos(240^\circ) = -\cos(60^\circ) = -\frac{1}{2}$.

- 32** Find $\tan(300^\circ)$ using the reference angle.

The reference angle is 60° ; in Quadrant IV, tangent is negative. $\tan(300^\circ) = -\tan(60^\circ) = -\sqrt{3}$.

Complementary angle relationships

- 33** Given $\sin(20^\circ) \approx 0.342$, use the relationship $\sin \theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ)$.

$$\cos(70^\circ) = \sin(20^\circ) \approx 0.342.$$

- 34** Given $\sin(35^\circ) \approx 0.574$, find $\cos(55^\circ)$.

$$\cos(55^\circ) = \sin(35^\circ) \approx 0.574.$$

The unit circle: coordinates of key points

- 35** State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 60^\circ$.

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

- 36** State the coordinates of the point on the unit circle at $\theta = 180^\circ$.

$$(-1, 0).$$

- 37** State the coordinates of the point on the unit circle at $\theta = 270^\circ$.

$$(0, -1).$$

Word problems: angles of elevation

- 38** This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35° .

a Write an equation relating the drone's height h to the given information, using the tangent ratio.
 $\tan(35^\circ) = \frac{h}{200}$, so $h = 200\tan(35^\circ)$.

b Find h , to the nearest meter, given $\tan(35^\circ) \approx 0.700$. $h \approx 200(0.700) = 140$ meters.

- 39 This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.
- a Write an equation for the height h the ladder reaches up the wall, using the sine ratio. $\sin(65^\circ) = \frac{h}{10}$, so $h = 10\sin(65^\circ)$.
- b Find h , to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$. $h \approx 10(0.906) = 9.1$ meters.
-

MCQ — Right-triangle trigonometric ratios

- 40 In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin \theta$.

- a. $\frac{17}{8}$
- b. $\frac{8}{15}$
- c. $\frac{15}{17}$
- d. $\frac{8}{17}$ ✓

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}.$$

- 41 In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos \theta$.

- a. $\frac{9}{41}$
- b. $\frac{9}{40}$
- c. $\frac{41}{40}$
- d. $\frac{40}{41}$ ✓

$$\text{Adjacent} = \sqrt{41^2 - 9^2} = 40; \cos \theta = \frac{40}{41}.$$

- 42 In a right triangle, angle θ has adjacent side 20 and hypotenuse 29. Find $\sin \theta$.

- a. $\frac{21}{20}$
- b. $\frac{20}{29}$
- c. $\frac{21}{29}$ ✓
- d. $\frac{29}{21}$

$$\text{Opposite} = \sqrt{29^2 - 20^2} = 21; \sin \theta = \frac{21}{29}.$$

- 43 In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin \theta$ and $\cos \theta$ (where θ is opposite the side of length 9).

- a. $\sin \theta = \frac{9}{40}, \cos \theta = \frac{40}{9}$
- b. $\sin \theta = \frac{40}{41}, \cos \theta = \frac{9}{41}$
- c. $\sin \theta = \frac{41}{9}, \cos \theta = \frac{41}{40}$
- d. $\sin \theta = \frac{9}{41}, \cos \theta = \frac{40}{41}$ ✓

Opposite = 9, adjacent = 40, hypotenuse = 41.

- 44 In a right triangle, angle θ has opposite side 7 and adjacent side 24. Find $\tan \theta$.

- a. $\frac{24}{7}$
- b. $\frac{24}{25}$
- c. $\frac{7}{25}$
- d. $\frac{7}{24}$ ✓

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{7}{24}.$$

MCQ — Special angles

- 45 State the exact value of $\cos(30^\circ)$.

- a. 1
- b. $\frac{\sqrt{3}}{2}$ ✓
- c. $\frac{\sqrt{2}}{2}$
- d. $\frac{1}{2}$

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}.$$

46 State the exact value of $\sin(45^\circ)$.

a. 1

b. $\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$ ✓

d. $\frac{1}{2}$

$$\sin(45^\circ) = \frac{\sqrt{2}}{2}.$$

47 State the exact value of $\tan(30^\circ)$.

a. $\frac{\sqrt{3}}{3}$ ✓

b. $\frac{1}{2}$

c. $\frac{\sqrt{3}}{2}$

d. $\sqrt{3}$

$$\tan(30^\circ) = \frac{\sin(30^\circ)}{\cos(30^\circ)} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

48 State the exact values of $\cos(90^\circ)$ and $\sin(0^\circ)$.

a. $\cos(90^\circ) = 0$, $\sin(0^\circ) = 1$

b. $\cos(90^\circ) = 1$, $\sin(0^\circ) = 0$

c. Both equal 1

d. Both equal 0 ✓

$$\cos(90^\circ) = 0 \text{ and } \sin(0^\circ) = 0.$$

49 State the exact value of $\tan(45^\circ)$.

a. Undefined

b. $\sqrt{2}$

c. 1 ✓

d. 0

$$\tan(45^\circ) = \frac{\sin(45^\circ)}{\cos(45^\circ)} = 1.$$

MCQ — Converting between radians and degrees

50 Convert 360° to radians.

- a. 4π
- b. π
- c. 2π ✓
- d. $\frac{\pi}{2}$

$$360 \times \frac{\pi}{180} = 2\pi.$$

51 Convert 45° to radians.

- a. $\frac{\pi}{6}$
- b. $\frac{\pi}{4}$ ✓
- c. $\frac{\pi}{2}$
- d. $\frac{\pi}{8}$

$$45 \times \frac{\pi}{180} = \frac{\pi}{4}.$$

52 Convert $\frac{3\pi}{4}$ radians to degrees.

- a. 120°
- b. 150°
- c. 135° ✓
- d. 270°

$$\frac{3\pi}{4} \times \frac{180}{\pi} = 135^\circ.$$

53 Convert 210° to radians.

- a. $\frac{7\pi}{12}$
- b. $\frac{7\pi}{3}$
- c. $\frac{7\pi}{6}$ ✓
- d. $\frac{5\pi}{6}$

$$210 \times \frac{\pi}{180} = \frac{7\pi}{6}.$$

MCQ — The Pythagorean identity

54 Given $\sin \theta = \frac{4}{5}$ with θ acute, find $\cos \theta$.

- a. $\frac{3}{5}$ ✓
- b. $\frac{4}{5}$
- c. $\frac{5}{3}$
- d. $\frac{9}{25}$

$$\cos^2 \theta = 1 - \frac{16}{25} = \frac{9}{25} \Rightarrow \cos \theta = \frac{3}{5}.$$

55 Given $\cos \theta = \frac{7}{25}$ with θ acute, find $\sin \theta$.

- a. $\frac{576}{625}$
- b. $\frac{24}{25}$ ✓
- c. $\frac{25}{24}$
- d. $\frac{7}{25}$

$$\sin^2 \theta = 1 - \frac{49}{625} = \frac{576}{625} \Rightarrow \sin \theta = \frac{24}{25}.$$

56 Given $\sin \theta = 0.8$, find $\cos^2 \theta$.

- a. 0.2
- b. 0.64
- c. 0.36 ✓
- d. 0.8

$$\cos^2 \theta = 1 - 0.8^2 = 1 - 0.64 = 0.36.$$

57 Given $\cos \theta = \frac{12}{13}$ with θ acute, find $\sin \theta$.

- a. $\frac{5}{13}$ ✓
- b. $\frac{13}{5}$
- c. $\frac{12}{13}$
- d. $\frac{25}{169}$

$$\sin^2 \theta = 1 - \frac{144}{169} = \frac{25}{169} \Rightarrow \sin \theta = \frac{5}{13}.$$

MCQ — Simplifying trigonometric expressions

58 Simplify $\sin^2 \theta + \cos^2 \theta - 1$.

- a. $2\sin^2 \theta$
- b. 0 ✓
- c. 2
- d. 1

By the Pythagorean identity, $\sin^2 \theta + \cos^2 \theta = 1$, so this is 0.

59 Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$ (for $\cos \theta \neq 0$).

a. $\frac{1}{\cos \theta}$

b. $\cos \theta$ ✓

c. $1 - \cos \theta$

d. $\sin \theta$

$1 - \sin^2 \theta = \cos^2 \theta$, so $\frac{\cos^2 \theta}{\cos \theta} = \cos \theta$.

60 Simplify $\frac{\sin \theta}{\tan \theta}$ (for $\tan \theta \neq 0$).

a. 1

b. $\sin \theta$

c. $\cos \theta$ ✓

d. $\frac{1}{\cos \theta}$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$, so $\frac{\sin \theta}{\sin \theta / \cos \theta} = \cos \theta$.

61 Simplify $2\sin^2 \theta + 2\cos^2 \theta$.

a. 1

b. 0

c. 2 ✓

d. $4\sin \theta \cos \theta$

$2(\sin^2 \theta + \cos^2 \theta) = 2(1) = 2$.

MCQ — Graphs: amplitude and period

62 State the amplitude and period of $y = 4\sin(x)$.

- a. Amplitude 2π , period 4
- b. Amplitude 4, period 2π ✓**
- c. Amplitude 1, period 4
- d. Amplitude 4, period π

For $y = A\sin(Bx)$ with $B = 1$: amplitude = 4, period = $\frac{2\pi}{1} = 2\pi$.

63 State the amplitude and period of $y = 2\cos(3x)$.

- a. Amplitude 2, period $\frac{2\pi}{3}$ ✓**
- b. Amplitude 3, period $\frac{2\pi}{2}$
- c. Amplitude 6, period 2π
- d. Amplitude 2, period 6π

Amplitude = 2; period = $\frac{2\pi}{3}$.

64 State the amplitude and period of $y = 5\sin\left(\frac{x}{2}\right)$.

- a. Amplitude 5, period π
- b. Amplitude 5, period 4π ✓**
- c. Amplitude $\frac{5}{2}$, period 2π
- d. Amplitude 2, period 4π

$B = \frac{1}{2}$: period = $\frac{2\pi}{1/2} = 4\pi$; amplitude = 5.

65 A cosine graph has amplitude 3 and period 4π . Write its equation in the form $y = A\cos(Bx)$.

- a. $y = 3\cos(4\pi x)$
- b. $y = 4\pi\cos(3x)$
- c. $y = 3\cos\left(\frac{x}{2}\right)$ ✓**
- d. $y = 3\cos(2x)$

$B = \frac{2\pi}{\text{period}} = \frac{2\pi}{4\pi} = \frac{1}{2}$.

66 The graph shows $y = A\sin(x)$ for some A . Find A by reading the graph's amplitude.

- a. $A = 1$
- b. $A = 2\pi$
- c. $A = 4$
- d. $A = 2$ ✓

The curve's peak height (amplitude) is 2.

MCQ — Graphs: phase shift and vertical shift

67 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the right and 3 units up. Write the new equation.

- a. $y = \cos(x) + \frac{\pi}{4} + 3$
- b. $y = \cos\left(x - \frac{\pi}{4}\right) + 3$ ✓
- c. $y = \cos\left(x + \frac{\pi}{4}\right) + 3$
- d. $y = \cos\left(x - \frac{\pi}{4}\right) - 3$

Right shift: replace x with $x - \frac{\pi}{4}$; up 3: add 3.

68 State the vertical shift and midline of $y = 2\sin(x) - 4$.

- a. Down 4, midline $y = 2$
- b. Up 4, midline $y = 4$
- c. Down 4, midline $y = -4$ ✓
- d. Down 2, midline $y = -4$

Subtracting 4 shifts down 4; the midline is $y = -4$.

69 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{6}$ to the left. Write the new equation.

- a. $y = \sin\left(\frac{\pi}{6} - x\right)$
- b. $y = \sin(x) + \frac{\pi}{6}$
- c. $y = \sin\left(x + \frac{\pi}{6}\right)$ ✓
- d. $y = \sin\left(x - \frac{\pi}{6}\right)$

Left shift: replace x with $x + \frac{\pi}{6}$.

70 The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.

- a. $y = 0$
- b. $y = 2$ ✓
- c. $y = 1$
- d. $y = -2$

The curve oscillates evenly about $y = 2$.

MCQ — Solving basic trigonometric equations

71 Solve $2\sin\theta = \sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 45^\circ$ only
- b. $\theta = 45^\circ$ or $\theta = 135^\circ$ ✓
- c. $\theta = 45^\circ$ or $\theta = 225^\circ$
- d. $\theta = 60^\circ$ or $\theta = 120^\circ$

$$\sin\theta = \frac{\sqrt{2}}{2}: \theta = 45^\circ \text{ or } 135^\circ.$$

72 Solve $\sqrt{3}\tan\theta = 3$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 60^\circ$ or $\theta = 300^\circ$

b. $\theta = 30^\circ$ or $\theta = 210^\circ$

c. $\theta = 60^\circ$ or $\theta = 240^\circ$ ✓

d. $\theta = 60^\circ$ only

$\tan\theta = \sqrt{3}$: $\theta = 60^\circ$ or 240° (tangent positive in Q1 and Q3).

73 Solve $2\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 30^\circ$ or $\theta = 330^\circ$

b. $\theta = 60^\circ$ or $\theta = 240^\circ$

c. $\theta = 60^\circ$ only

d. $\theta = 60^\circ$ or $\theta = 300^\circ$ ✓

$\cos\theta = 0.5$: $\theta = 60^\circ$ or 300° .

74 Solve $\tan\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 45^\circ$ or $\theta = 225^\circ$ ✓

b. $\theta = 45^\circ$ only

c. $\theta = 45^\circ$ or $\theta = 315^\circ$

d. $\theta = 45^\circ$ or $\theta = 135^\circ$

Tangent is positive in Q1 and Q3: $\theta = 45^\circ$ or 225° .

75 Solve $2\sin\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 210^\circ$ or $\theta = 300^\circ$

b. $\theta = 150^\circ$ or $\theta = 330^\circ$

c. $\theta = 30^\circ$ or $\theta = 150^\circ$

d. $\theta = 210^\circ$ or $\theta = 330^\circ$ ✓

$\sin\theta = -0.5$: $\theta = 210^\circ$ or 330° .

MCQ — Reference angles in other quadrants

76 Find $\sin(120^\circ)$ using the reference angle.

- a. $\frac{\sqrt{3}}{2}$ ✓
- b. $-\frac{\sqrt{3}}{2}$
- c. $\frac{1}{2}$
- d. $-\frac{1}{2}$

Reference angle 60° ; sine is positive in Q2: $\sin(120^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$.

77 Find $\cos(210^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{3}}{2}$ ✓
- b. $\frac{\sqrt{3}}{2}$
- c. $-\frac{1}{2}$
- d. $\frac{1}{2}$

Reference angle 30° ; cosine is negative in Q3: $\cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$.

78 Find $\tan(330^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{3}}{3}$ ✓
- b. $\sqrt{3}$
- c. $-\sqrt{3}$
- d. $\frac{\sqrt{3}}{3}$

Reference angle 30° ; tangent is negative in Q4: $\tan(330^\circ) = -\tan(30^\circ) = -\frac{\sqrt{3}}{3}$.

79 Find $\sin(315^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{2}}{2}$ ✓
- b. $-\frac{1}{2}$
- c. $\frac{\sqrt{2}}{2}$
- d. $\frac{1}{2}$

Reference angle 45° ; sine is negative in Q4: $\sin(315^\circ) = -\sin(45^\circ) = -\frac{\sqrt{2}}{2}$.

MCQ — Complementary angle relationships

80 Given $\sin(25^\circ) \approx 0.423$, use the relationship $\sin \theta = \cos(90^\circ - \theta)$ to find $\cos(65^\circ)$.

- a. -0.423
- b. 0.906
- c. 0.577
- d. 0.423 ✓

$$\cos(65^\circ) = \cos(90^\circ - 25^\circ) = \sin(25^\circ) \approx 0.423.$$

81 Given $\cos(40^\circ) \approx 0.766$, find $\sin(50^\circ)$.

- a. 0.643
- b. 0.766 ✓
- c. 1.306
- d. 0.234

$$\sin(50^\circ) = \sin(90^\circ - 40^\circ) = \cos(40^\circ) \approx 0.766.$$

82 Given $\sin(70^\circ) \approx 0.940$, find $\cos(20^\circ)$.

- a. 1.064
- b. 0.060
- c. 0.342
- d. 0.940 ✓

$$\cos(20^\circ) = \cos(90^\circ - 70^\circ) = \sin(70^\circ) \approx 0.940.$$

MCQ — The unit circle: coordinates of key points

83 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 30^\circ$.

- a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- b. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
- c. $(1, 0)$
- d. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}, \sin(30^\circ) = \frac{1}{2}.$$

84 State the coordinates of the point on the unit circle at $\theta = 90^\circ$.

- a. $(1, 0)$
- b. $(0, -1)$
- c. $(0, 1)$ ✓
- d. $(-1, 0)$

$$\cos(90^\circ) = 0, \sin(90^\circ) = 1.$$

85 State the coordinates of the point on the unit circle at $\theta = 225^\circ$.

- a. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
- b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ ✓
- c. $(-1, -1)$
- d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

225° is in Q3, reference angle 45° : both coordinates negative, $-\frac{\sqrt{2}}{2}$.

86 The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?

- a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$
- b. $(0.5, 0.5)$
- c. $(1, 0)$
- d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$ ✓

$\cos(60^\circ) = 0.5$, $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.

MCQ — Word problems: angles of elevation

87 A drone hovers directly above a landmark. From a point 150 meters from the landmark's base, the angle of elevation to the drone is 40° . Find the drone's height, to the nearest meter.

- a. 150
- b. 196
- c. 126 ✓
- d. 115

$h = 150 \tan(40^\circ) \approx 126$ m.

- 88 A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.
- a. 4
 - b. 12
 - c. 8
 - d. 11 ✓

$$h = 12\sin(70^\circ) \approx 11 \text{ m.}$$

- 89 From the top of an 80 m cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest meter.
- a. 37
 - b. 80
 - c. 172 ✓
 - d. 189

$$\tan(25^\circ) = \frac{80}{d} \Rightarrow d = \frac{80}{\tan(25^\circ)} \approx 172 \text{ m.}$$

Multiple choice — easy

- 90 State the exact value of $\tan(45^\circ)$.
- a. $\frac{\sqrt{2}}{2}$
 - b. $\frac{\sqrt{3}}{2}$
 - c. 1 ✓
 - d. $\sqrt{3}$

$$\tan(45^\circ) = 1.$$

91 State the exact value of $\tan(0^\circ)$.

a. *undefined*

b. $\frac{\sqrt{3}}{2}$

c. 0 ✓

d. 1

$$\tan(0^\circ) = 0.$$

92 State the exact value of $\tan(30^\circ)$.

a. 0

b. $\frac{\sqrt{3}}{3}$ ✓

c. 1

d. $\frac{\sqrt{3}}{2}$

$$\tan(30^\circ) = \frac{\sqrt{3}}{3}.$$

93 State the exact value of $\sin(0^\circ)$.

a. $\frac{\sqrt{2}}{2}$

b. $\sqrt{3}$

c. $\frac{1}{2}$

d. 0 ✓

$$\sin(0^\circ) = 0.$$

94 State the exact value of $\sin(60^\circ)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{\sqrt{3}}{3}$

c. 1

d. $\sqrt{3}$

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}.$$

95 State the exact value of $\cos(30^\circ)$.

a. $\frac{1}{2}$

b. *undefined*

c. $\frac{\sqrt{3}}{2}$ ✓

d. $\sqrt{3}$

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}.$$

96 State the exact value of $\sin(90^\circ)$.

a. 0

b. 1 ✓

c. *undefined*

d. $\frac{\sqrt{3}}{3}$

$$\sin(90^\circ) = 1.$$

97 State the exact value of $\sin(45^\circ)$.

a. $\frac{\sqrt{3}}{2}$

b. *undefined*

c. $\frac{\sqrt{2}}{2}$ ✓

d. $\sqrt{3}$

$$\sin(45^\circ) = \frac{\sqrt{2}}{2}.$$

98 In a right triangle, angle θ has opposite side 18, adjacent side 80, and hypotenuse 82. Find $\sin \theta$.

a. $\frac{41}{9}$

b. $\frac{3}{14}$

c. $\frac{8}{41}$

d. $\frac{9}{41}$ ✓

$$\sin \theta = \frac{18}{82} = \frac{9}{41}.$$

99 In a right triangle, angle θ has opposite side 80, adjacent side 18, and hypotenuse 82. Find $\tan \theta$.

a. $\frac{40}{9}$ ✓

b. 4

c. $\frac{13}{3}$

d. $\frac{9}{40}$

$$\tan \theta = \frac{80}{18} = \frac{40}{9}.$$

100 In a right triangle, angle θ has opposite side 15, adjacent side 8, and hypotenuse 17. Find $\tan \theta$.

a. $\frac{3}{2}$

b. $\frac{8}{15}$

c. $\frac{15}{8}$ ✓

d. $\frac{13}{8}$

$$\tan \theta = \frac{15}{8}.$$

101 In a right triangle, angle θ has opposite side 24, adjacent side 7, and hypotenuse 25. Find $\tan \theta$.

a. $\frac{7}{24}$

b. $\frac{8}{3}$

c. $\frac{22}{7}$

d. $\frac{24}{7}$ ✓

$$\tan \theta = \frac{24}{7}.$$

102 In a right triangle, angle θ has opposite side 16, adjacent side 30, and hypotenuse 34. Find $\sin \theta$.

a. $\frac{7}{17}$

b. $\frac{17}{8}$

c. $\frac{4}{9}$

d. $\frac{8}{17}$ ✓

$$\sin \theta = \frac{16}{34} = \frac{8}{17}.$$

103 In a right triangle, angle θ has opposite side 80, adjacent side 18, and hypotenuse 82. Find $\sin \theta$.

a. $\frac{39}{41}$

b. $\frac{41}{40}$

c. $\frac{40}{41}$ ✓

d. $\frac{20}{21}$

$$\sin \theta = \frac{80}{82} = \frac{40}{41}.$$

104 In a right triangle, angle θ has opposite side 48, adjacent side 14, and hypotenuse 50. Find $\tan \theta$.

a. 3

b. $\frac{24}{7}$ ✓

c. $\frac{23}{7}$

d. $\frac{7}{24}$

$$\tan \theta = \frac{48}{14} = \frac{24}{7}.$$

Multiple choice — medium

105 Given $\sin \theta = \frac{9}{41}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{40}{41}$ ✓

b. $\frac{1}{41}$

c. $\frac{38}{41}$

d. $\frac{42}{41}$

$$\cos^2 \theta = 1 - \frac{81}{1681} = \frac{1600}{1681}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{40}{41}.$$

106 Given $\sin \theta = \frac{7}{25}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{26}{25}$

b. $\frac{22}{25}$

c. $\frac{24}{25}$ ✓

d. $\frac{1}{25}$

$$\cos^2 \theta = 1 - \frac{49}{625} = \frac{576}{625}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{24}{25}.$$

107 Given $\sin \theta = \frac{12}{37}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{2}{37}$

b. $\frac{33}{37}$

c. $\frac{35}{37}$ ✓

d. 1

$$\cos^2 \theta = 1 - \frac{144}{1369} = \frac{1225}{1369}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{35}{37}.$$

108 Given $\sin \theta = \frac{5}{13}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{1}{13}$

b. $\frac{14}{13}$

c. $\frac{10}{13}$

d. $\frac{12}{13}$ ✓

$$\cos^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{12}{13}.$$

109 Given $\cos \theta = \frac{7}{25}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{26}{25}$

b. $\frac{22}{25}$

c. $\frac{1}{25}$

d. $\frac{24}{25}$ ✓

$$\sin^2 \theta = 1 - \frac{49}{625} = \frac{576}{625}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{24}{25}.$$

110 Convert $\frac{\pi}{4}$ radians to degrees.

a. 15°

b. 45° ✓

c. 105°

d. 75°

$$\frac{\pi}{4} \times \frac{180^\circ}{\pi} = 45^\circ.$$

111 Convert $\frac{5\pi}{6}$ radians to degrees.

a. 150° ✓

b. 120°

c. 180°

d. 210°

$$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ.$$

112 Convert 90° to radians.

- a. $\frac{2\pi}{3}$
- b. $\frac{\pi}{2}$ ✓
- c. $\frac{\pi}{3}$
- d. $\frac{5\pi}{6}$

$$90^\circ = \frac{\pi}{2} \text{ radians.}$$

113 Convert $\frac{3\pi}{2}$ radians to degrees.

- a. 330°
- b. 270° ✓
- c. 240°
- d. 300°

$$\frac{3\pi}{2} \times \frac{180^\circ}{\pi} = 270^\circ.$$

114 Convert 180° to radians.

- a. $\frac{4\pi}{3}$
- b. $\frac{7\pi}{6}$
- c. $\frac{5\pi}{6}$
- d. π ✓

$$180^\circ = \pi \text{ radians.}$$

115 State the amplitude and period of $y = 2\cos(4x)$.

- a. Amplitude: 3. Period: $\frac{\pi}{2}$
- b. Amplitude: 2. Period: $\frac{2\pi}{5}$
- c. Amplitude: 3. Period: $\frac{2\pi}{5}$
- d. Amplitude: 2. Period: $\frac{\pi}{2}$ ✓

$$\text{Amplitude: 2. Period: } \frac{2\pi}{4} = \frac{\pi}{2}.$$

116 State the amplitude and period of $y = 4\sin(4x)$.

a. Amplitude: 4. Period: $\frac{\pi}{2}$ ✓

b. Amplitude: 4. Period: $\frac{2\pi}{5}$

c. Amplitude: 5. Period: $\frac{2\pi}{5}$

d. Amplitude: 5. Period: $\frac{\pi}{2}$

Amplitude: 4. Period: $\frac{2\pi}{4} = \frac{\pi}{2}$.

117 State the amplitude and period of $y = 6\cos(2x)$.

a. Amplitude: 7. Period: π

b. Amplitude: 6. Period: π ✓

c. Amplitude: 7. Period: $\frac{2\pi}{3}$

d. Amplitude: 6. Period: $\frac{2\pi}{3}$

Amplitude: 6. Period: $\frac{2\pi}{2} = \pi$.

118 State the amplitude and period of $y = 6\sin(3x)$.

a. Amplitude: 7. Period: $\frac{\pi}{2}$

b. Amplitude: 6. Period: $\frac{\pi}{2}$

c. Amplitude: 7. Period: $\frac{2\pi}{3}$

d. Amplitude: 6. Period: $\frac{2\pi}{3}$ ✓

Amplitude: 6. Period: $\frac{2\pi}{3}$.

119 State the amplitude and period of $y = 5\cos(4x)$.

- a. Amplitude: 5. Period: $\frac{2\pi}{5}$
 - b. Amplitude: 6. Period: $\frac{2\pi}{5}$
 - c. Amplitude: 6. Period: $\frac{\pi}{2}$
 - d. Amplitude: 5. Period: $\frac{\pi}{2}$ ✓
- Amplitude: 5. Period: $\frac{2\pi}{4} = \frac{\pi}{2}$.

Multiple choice — hard

120 Solve $2\sin \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 210^\circ$ or $\theta = 330^\circ$ ✓
- b. $\theta = 210^\circ$ or $\theta = 60^\circ$
- c. $\theta = 150^\circ$ or $\theta = 30^\circ$
- d. $\theta = 300^\circ$ or $\theta = 330^\circ$

$\sin \theta = -\frac{1}{2}$. Solutions: $\theta = 210^\circ$ or $\theta = 330^\circ$.

121 Solve $\tan \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 225^\circ$ or $\theta = 315^\circ$
- b. $\theta = 135^\circ$ or $\theta = 315^\circ$ ✓
- c. $\theta = 225^\circ$ or $\theta = 45^\circ$
- d. $\theta = 135^\circ$ or $\theta = 45^\circ$

Solutions: $\theta = 135^\circ$ or $\theta = 315^\circ$.

122 Solve $\sqrt{2} \cos \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 225^\circ$ or $\theta = 135^\circ$

b. $\theta = 135^\circ$ or $\theta = 315^\circ$

c. $\theta = 135^\circ$ or $\theta = 225^\circ$ ✓

d. $\theta = 145^\circ$ or $\theta = 235^\circ$

$\cos \theta = -\frac{\sqrt{2}}{2}$. Solutions: $\theta = 135^\circ$ or $\theta = 225^\circ$.

123 Solve $\sqrt{2} \cos \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 55^\circ$ or $\theta = 325^\circ$

b. $\theta = 45^\circ$ or $\theta = 315^\circ$ ✓

c. $\theta = 315^\circ$ or $\theta = 45^\circ$

d. $\theta = 135^\circ$ or $\theta = 315^\circ$

$\cos \theta = \frac{\sqrt{2}}{2}$. Solutions: $\theta = 45^\circ$ or $\theta = 315^\circ$.

124 Solve $2 \cos \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 120^\circ$ or $\theta = 240^\circ$ ✓

b. $\theta = 120^\circ$ or $\theta = 330^\circ$

c. $\theta = 240^\circ$ or $\theta = 120^\circ$

d. $\theta = 210^\circ$ or $\theta = 240^\circ$

$\cos \theta = -\frac{1}{2}$. Solutions: $\theta = 120^\circ$ or $\theta = 240^\circ$.

125 Find $\sin(330^\circ)$ using the reference angle.

a. $-\frac{1}{2}$ ✓

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{1}{2}$

d. $-\frac{\sqrt{3}}{3}$

The reference angle is 30° ; in Quadrant IV, sine is negative. $\sin(330^\circ) = -\sin(30^\circ) = -\frac{1}{2}$.

126 Find $\sin(150^\circ)$ using the reference angle.

- a. $\frac{1}{2}$ ✓
- b. $\frac{\sqrt{3}}{2}$
- c. $-\frac{1}{2}$
- d. $\frac{\sqrt{3}}{3}$

The reference angle is 30° ; in Quadrant II, sine is positive. $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$.

127 Find $\cos(135^\circ)$ using the reference angle.

- a. 1
- b. $\frac{\sqrt{2}}{2}$
- c. -1
- d. $-\frac{\sqrt{2}}{2}$ ✓

The reference angle is 45° ; in Quadrant II, cosine is negative. $\cos(135^\circ) = -\cos(45^\circ) = -\frac{\sqrt{2}}{2}$.

128 Find $\sin(210^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{3}}{3}$
- b. $\frac{1}{2}$
- c. $-\frac{1}{2}$ ✓
- d. $-\frac{\sqrt{3}}{2}$

The reference angle is 30° ; in Quadrant III, sine is negative. $\sin(210^\circ) = -\sin(30^\circ) = -\frac{1}{2}$.

129 Find $\sin(120^\circ)$ using the reference angle.

- a. $\frac{1}{2}$
- b. $\frac{\sqrt{3}}{2}$ ✓
- c. $\sqrt{3}$
- d. $-\frac{\sqrt{3}}{2}$

The reference angle is 60° ; in Quadrant II, sine is positive. $\sin(120^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$.

130 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 60^\circ$.

- a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ✓
- b. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
- c. $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
- d. $(1, 0)$

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

131 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 30^\circ$.

- a. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓
- b. $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
- c. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
- d. $(0, 1)$

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right).$$

132 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 315^\circ$.

- a. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
 - b. $(-1, 0)$
 - c. $(0, 1)$
 - d. $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ ✓
- $$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

133 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 330^\circ$.

a. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

b. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

d. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ ✓

$\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$.

134 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 300^\circ$.

a. $(1, 0)$

b. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ ✓

c. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

d. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

$\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$.